

Activity 10 - Diagnose the Pilot Quality Failure

Course: Effective Project Management for Small Projects (TGS-2020505545) Topic: 3 | Apply Methods and Tools to Drive a Project Learning outcome: LO3 Duration: 40 minutes Team size: 3-4 learners Tool set: Check Sheet, Fishbone, Five Whys, Pareto and Control Chart

Objective

Use evidence and root-cause tools to select a controlled corrective experiment (A3, K3).

The situation

During testing, only 82% of orders are ready on time and pick accuracy is 94%. The vendor blames store staff; store staff blame inventory updates. The sponsor wants action before facts and hypotheses are separated.

What you will do

Reconcile defect data, broaden possible causes, test a causal chain, prioritise the vital few, interpret process behaviour and design one measurable corrective experiment.

What you will produce

A check sheet, fishbone, five-whys chain, Pareto chart, control-chart interpretation and SMART corrective experiment.

Tool files in this folder

- [`tools/35 Check Sheets.xlsx`](#)
- [`tools/37 Ishikawa Diagram.xlsx`](#)
- [`tools/38 Five Whys.xlsx`](#)
- [`tools/36 Pareto Principle Template.xlsx`](#)
- [`tools/34 Control Chart.xlsx`](#)

Data files

- [`data/quality-test-data.csv`](#)

Step-by-step instructions

1. Enter and reconcile the supplied defect data by type, outlet, shift and date.
2. Build a fishbone across People, Process, Technology, Data, Measurement and Environment; label facts and hypotheses.
3. Select a high-frequency cause family and complete the five-whys chain to an influenceable system cause.
4. Read the chain backwards with because and repair causal jumps.
5. Rank defects in the Pareto template, calculate cumulative percentage and identify the vital few.
6. Plot daily performance and distinguish control behaviour from the customer specification.
7. Create one SMART corrective experiment with owner, measure, due date and review rule.

Evidence to submit

Complete [WORKSHEET.md](#) and attach/export the completed tool files named above.

Acceptance criteria

Data totals reconcile, Pareto reaches 100%, the causal chain is actionable, the control interpretation matches the signal and the experiment has an owner and verification date.

Debrief

- Fishbone broadens, Five Whys deepens and Pareto prioritises.
- Control limits describe process behaviour; specifications describe customer need.
- Improvement is project work only when funded, owned and verified.

The full procedure is also reproduced in the Learner Guide, Activity 10. The trainer deck contains only the briefing and success criteria.